

Oral Cavity in Disease



3rd BDS / Chapter - 3: Dental Caries

Dr. Syed Mukith

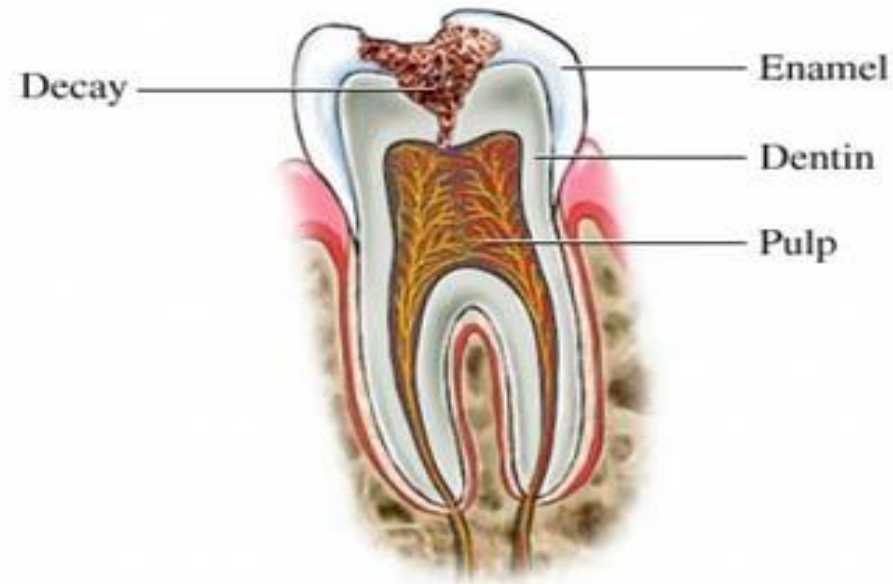
Learning Outcomes

- **By the end of this chapter the students will be able to:**
 1. Describe Etiopathogenesis of Dental Caries.
 2. Classify the carious lesions of the teeth.
 3. Explain the appearance of Dental Caries in Microscopy and zones of Enamel and Dentin Caries.
 4. Explain the method of prevention of Dental Caries and treatment of Dental Caries.
 5. Illustrate the sequelae of dental caries.

Etiopathogenesis of Dental Caries

INTRODUCTION

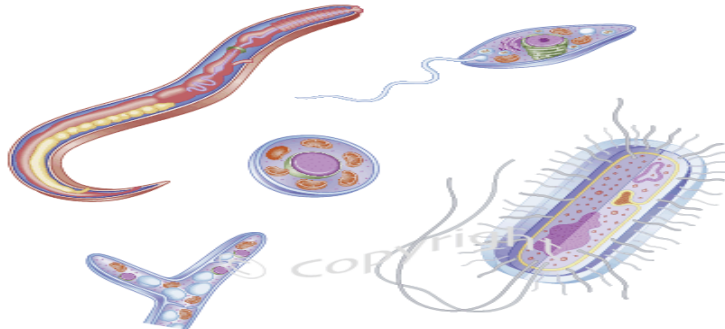
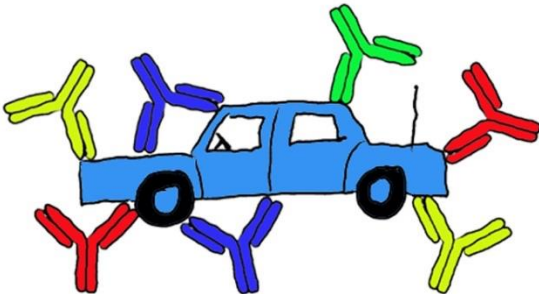
Dental Caries is a microbial disease of the calcified tissues of teeth, characterized by demineralization of the inorganic portion and destruction of the organic substance of the tooth

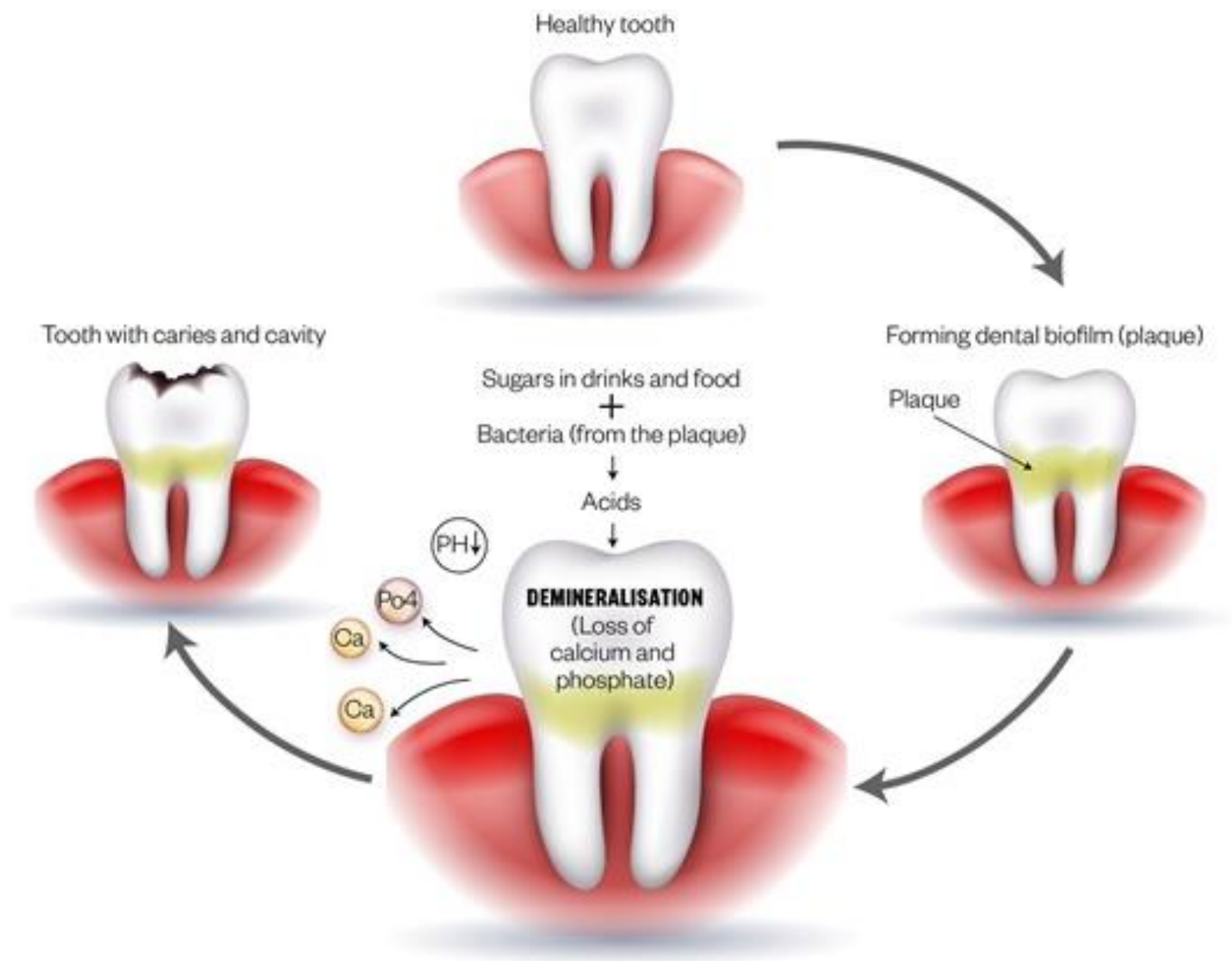


ETIOLOGY/THEORIES

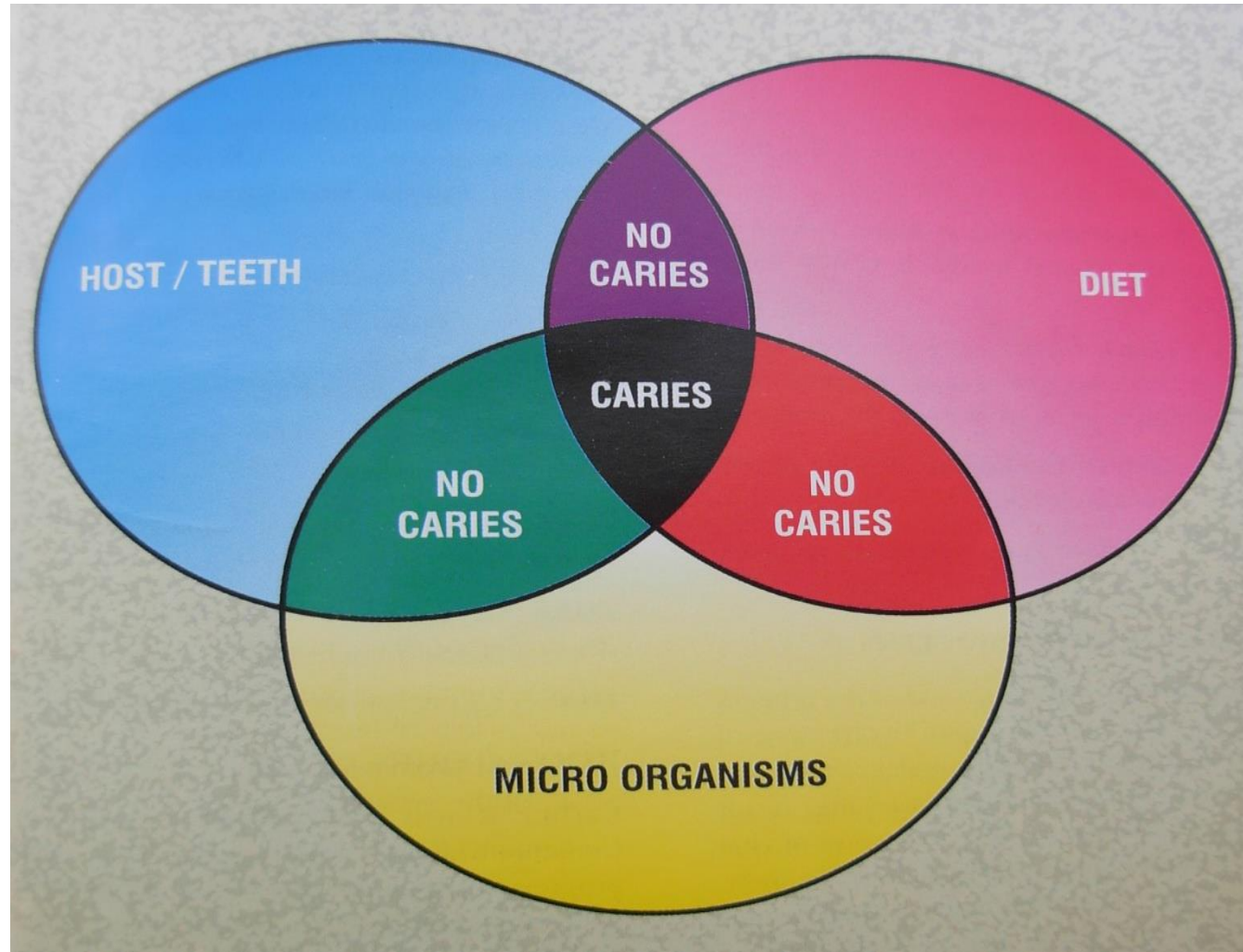


1. Worms Theory
2. Humors Theory
3. Vital Theory
4. Chemical Theory
5. Parasitic Theory
6. Auto Immunity
7. Proteolytic – Chelation Theory
8. Acidogenic Theory





Acidogenic Theory: Keyes Triad

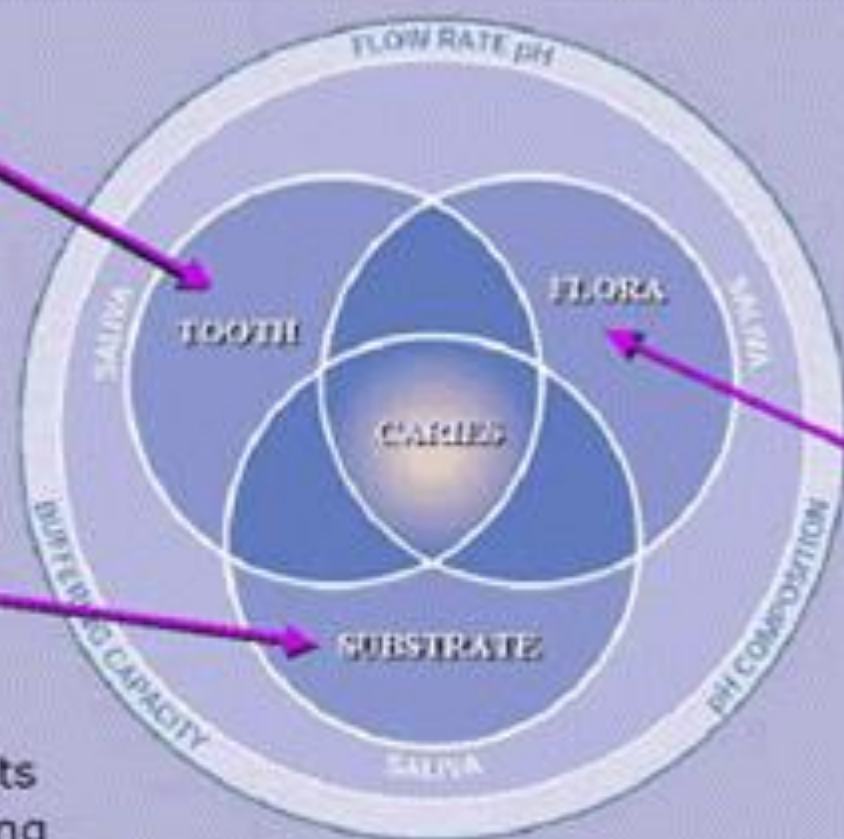


TOOTH

Age
Fluorides
Morphology
Nutrition
Trace Elements
Carbonate Level

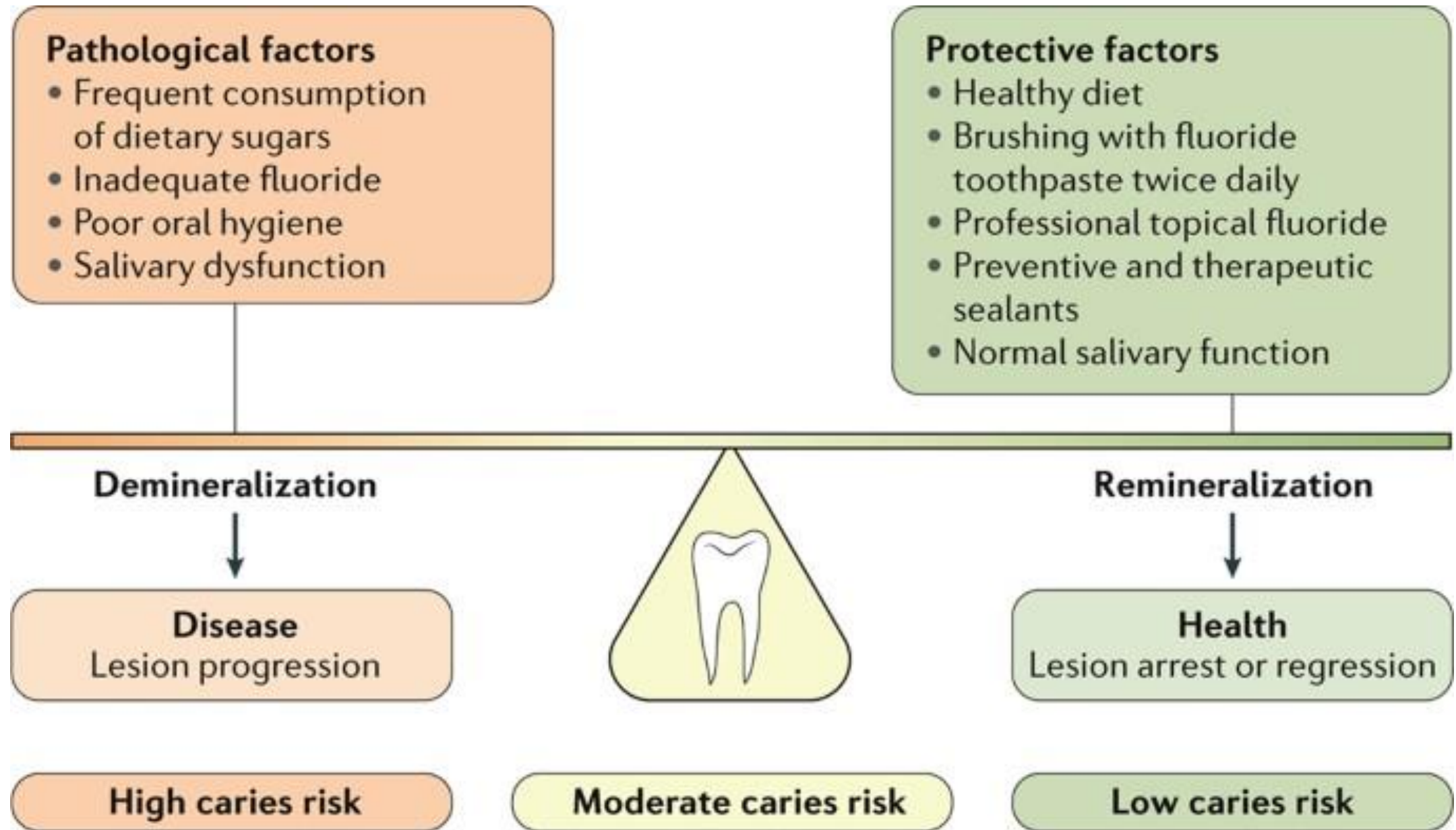
SUBSTRATE

Oral Clearance
Oral Hygiene
Salivary Stimulants
Frequency of Eating
Carbohydrate (type, concentration)



FLORA

Strep, Mutans
(Substrate)
Oral Hygiene
Fluoride in
Plaque



FACTORS FOR DENTAL CARIES

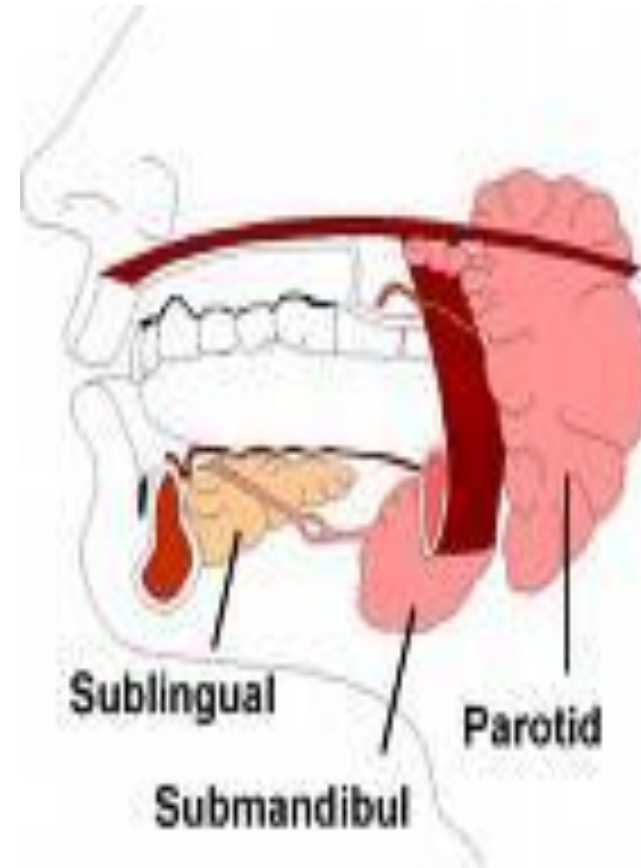
HOST FACTORS

- **Tooth**

1. Composition.
2. Morphologic characteristics.
3. Position.

- **Saliva**

1. Composition.
 1. Inorganic.
 2. Organic.
2. pH.
3. Quantity.
4. Viscosity.
5. Antibacterial factors.



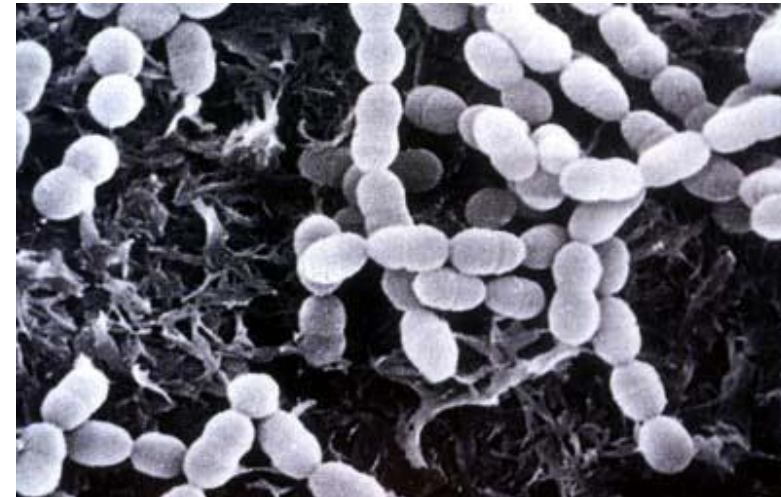
DIET

1. Physical factors.
 1. Quality of diet.
2. Local factors.
 1. Carbohydrate content.
 2. Fluorine content.



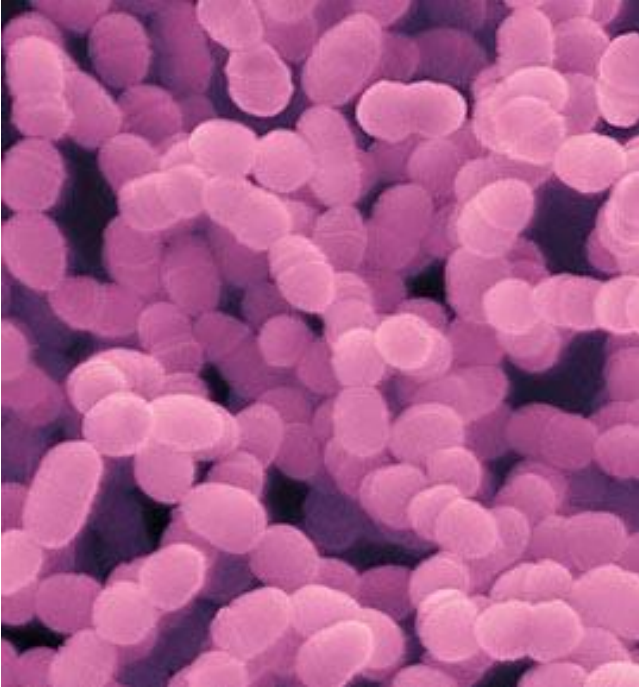
MICRO ORGANISMS

1. Streptococcus mutans.
2. Oral Lactobacilli
3. Oral Actinomyces

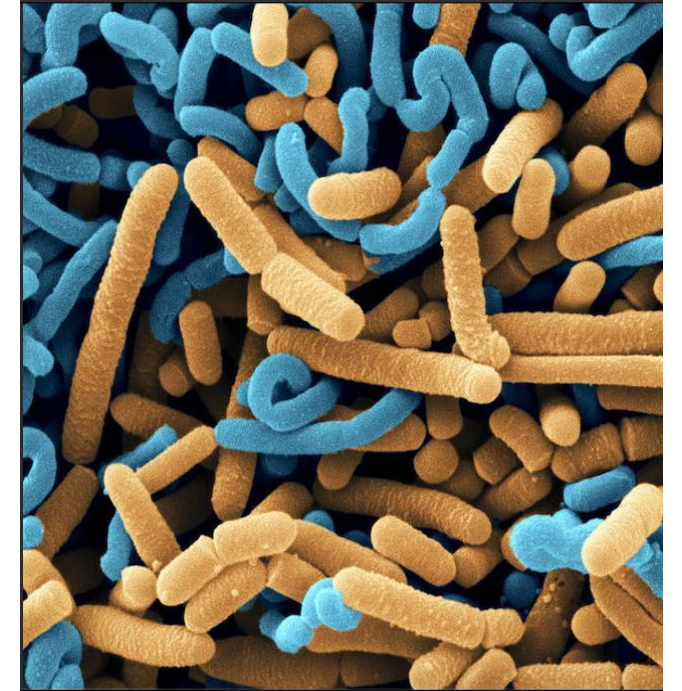
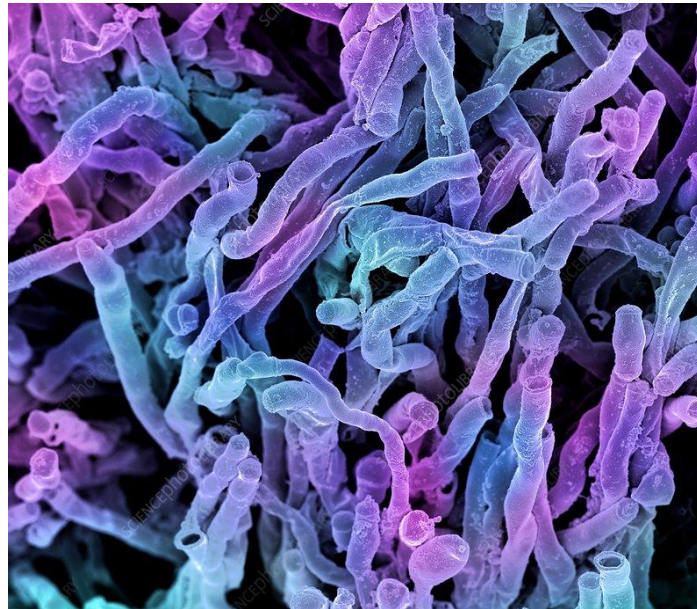


Microbiology of Dental Caries

Microorganisms



- Streptococcus Mutants
- Lactobacillus
- Actinomyces



Metabolism & Nutrition

- ability – attach to the tooth surface and form a biofilm – produces slime, divides and produces micro-colonies within the slime layer, and constructs a biofilm – resistant to detergent and biocides.
- Heterotrophs.

Ecology

may affect other plaque bacteria in the mouth.

Roles

- Dental caries (Tooth decay) - an oral infectious disease.
- *S. mutans* metabolize simple sugars and other fermentable carbohydrates into acids – dissolve tooth enamel.

Streptococcus mutans

Classification

Domain: Bacteria
Phylum: Firmicutes
Class: Bacilli
Order: Lactobacillales
Family: Streptococcaceae
Genus: *Streptococcus*
Species: *mutans*

Description

- first described by James Kilian Clarke (1886-1950)
- gram-positive.
 - cocci shaped bacteria.
 - facultative anaerobes.
 - mesophilic.
 - produce asexually.
 - found in human oral cavity.



- **ORAL LACTOBACILLI**

- Lactobacilli are *gram positive ,non spore –forming rods* that generally grow best under micro aerophilic conditions. Isolation and enumeration of oral lactobacilli have been facilitated by use of selective agar medium which suppresses the growth of other oral organisms by its low pH (5.4) .
- Lactobacilli are found mostly as transients in the mouth of infants . Lactobacilli represents about 1% of the oral flora. *L. casei* and *L fermentum* are the most common oral species. The population of oral lactobacilli is influenced by dietary habits. A favorite habitat of lactobacilli is in the dentin of deep carious lesions.

Oral actinomyces

Actinomyces is a *gram-positive, nonmotile, non-spore-forming* organisms occurring as rods and filaments that vary considering in length . Filaments are usually long and slender and are branching .

The species that have been found in the oral cavity are:

Facultative anaerobic

A. naeslundii

A. viscosus

anaerobic

A. israelii

Actinomyces meyeri

A. odontolyticus

Classification of Dental Caries

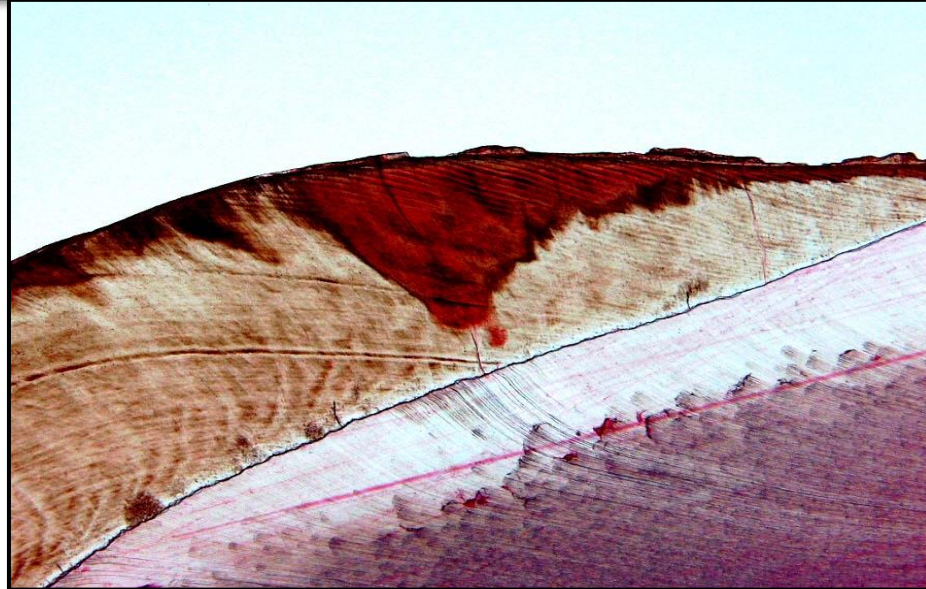
Classification Based on the Following:

- 1. Morphology of teeth**
- 2. Age**
- 3. Progression**
- 4. Surfaces Involved**

1.1 PIT & FISSURE CARIES



1.2 SMOOTH SURFACE CARRIES



1.3 ROOT CARIES



2. According to Age

2.1 Nursing bottle caries

2.2 Adolescent caries

2.3 Geriatric caries/Senile caries



3. According to Severity and Progress of lesion

3.1 Incipient Caries/Initial/Reversible Caries

3.2 Rampant Caries

3.3 Arrested Caries/Eburnation

3.4 Recurrent Caries/Secondary Caries

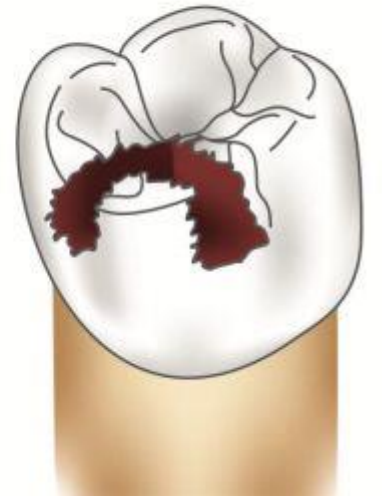
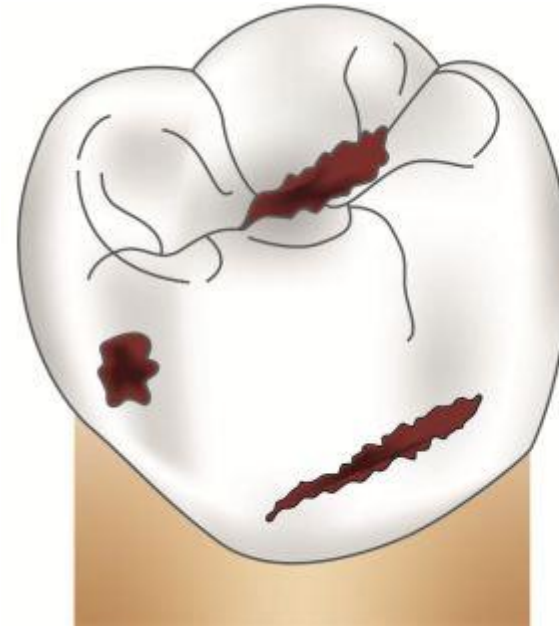
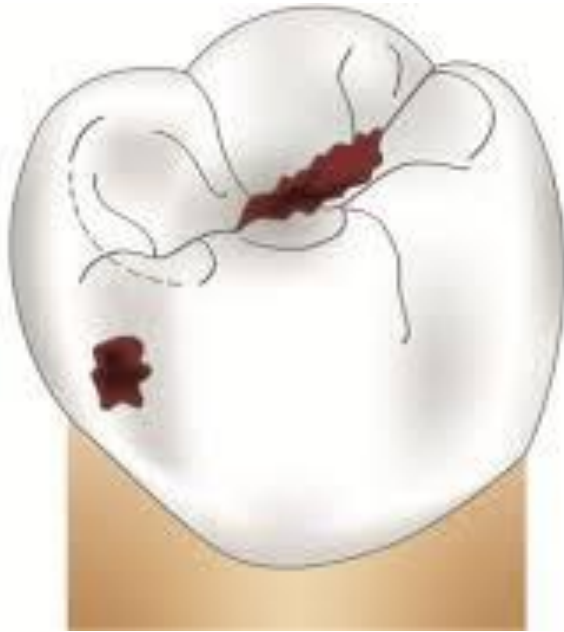


4. According to surfaces involved

4.1 Simple

4.2 Compound

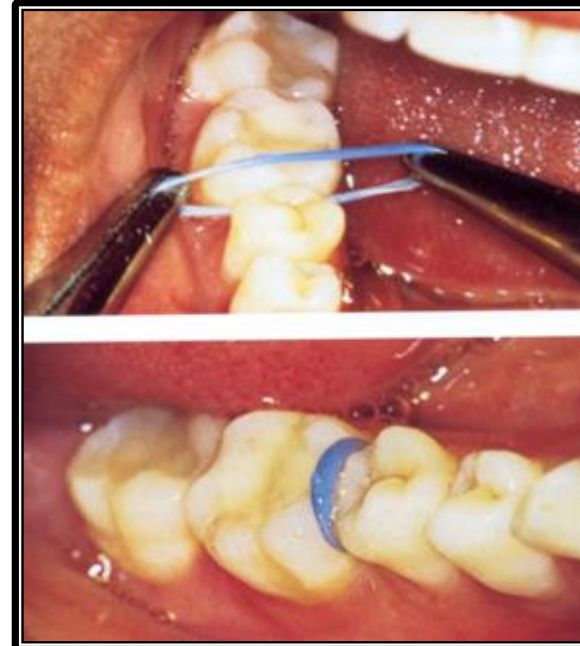
4.3 Complex



Diagnosis of Dental Caries

DIAGNOSIS OF DENTAL CARIES

1. Visual examination.
2. Visual and Tactile examination.
3. Enhanced visual examination.
4. Dye penetration method



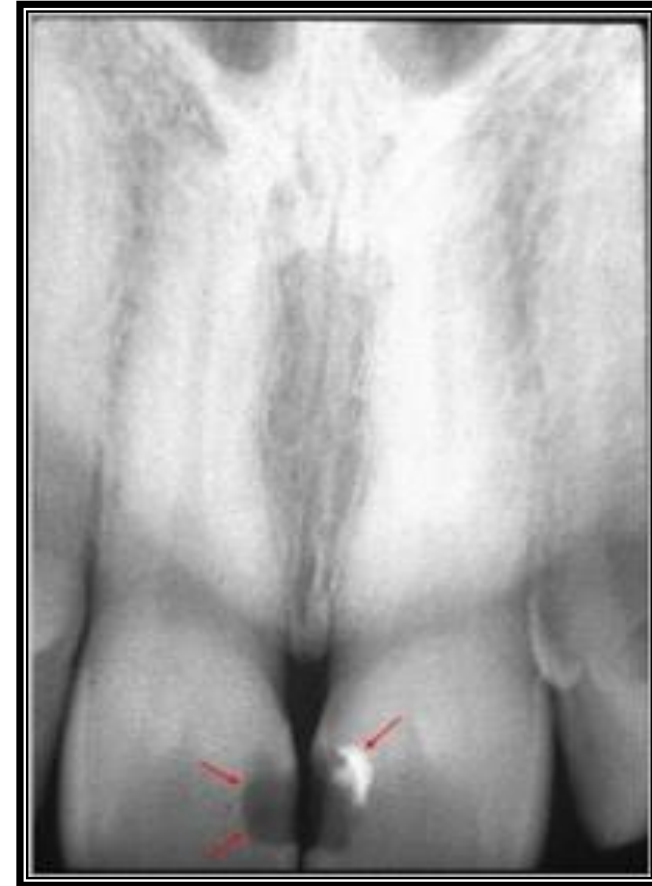
5. Radiographic examination

Conventional

Xero radiography

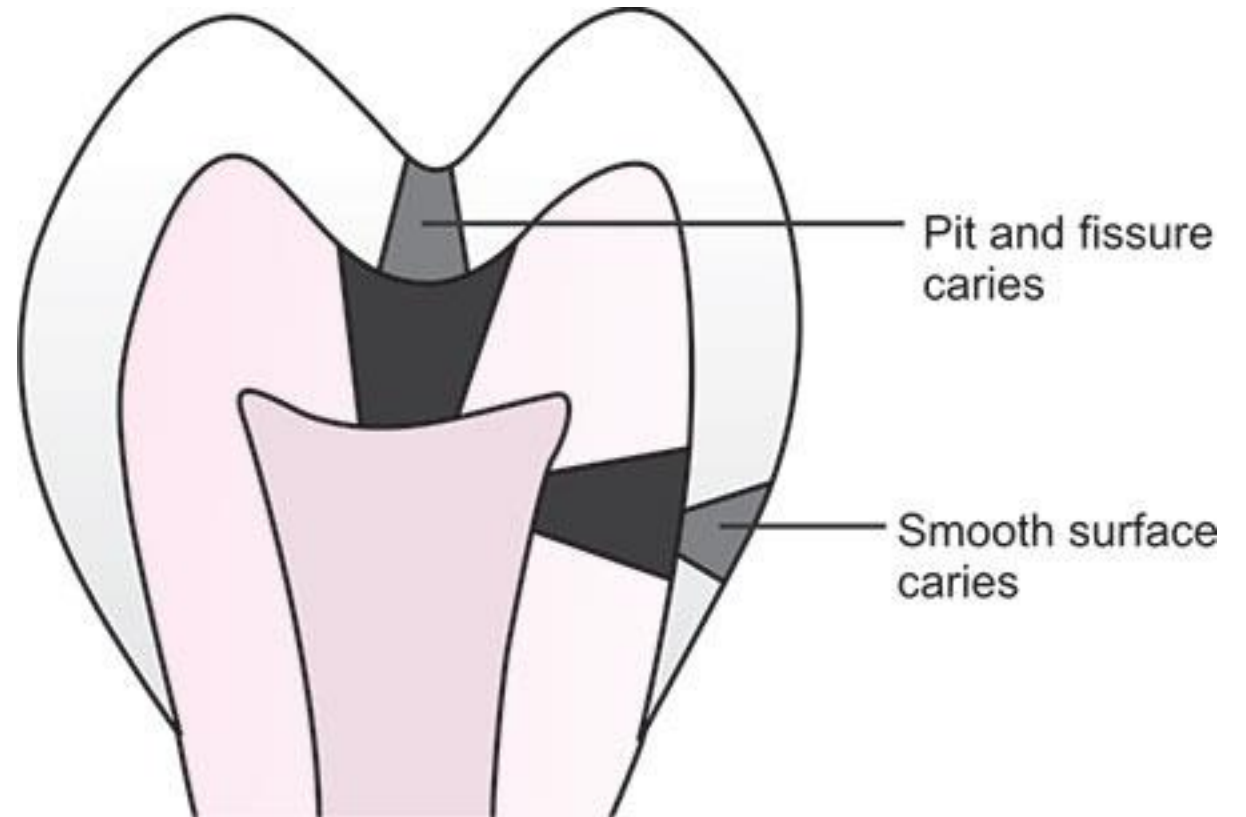
Digital imaging

6. Fibro optic method

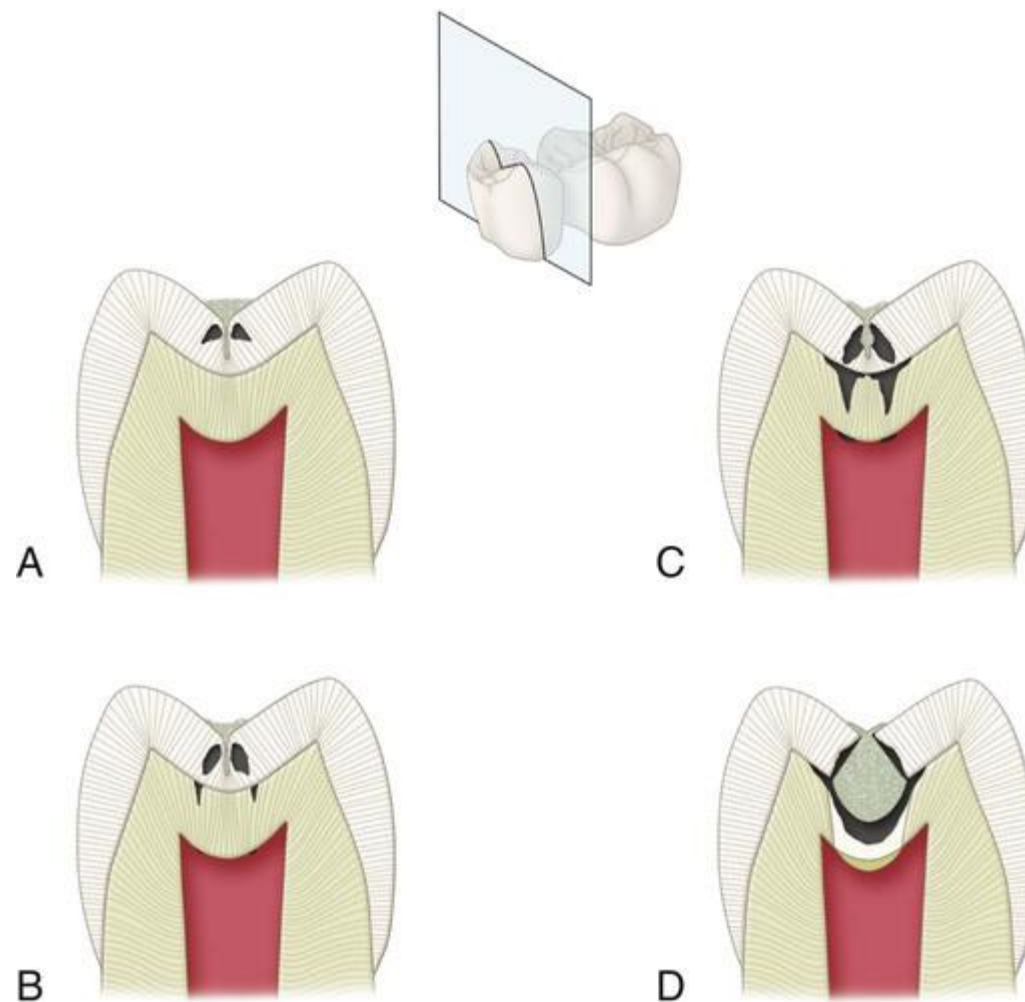
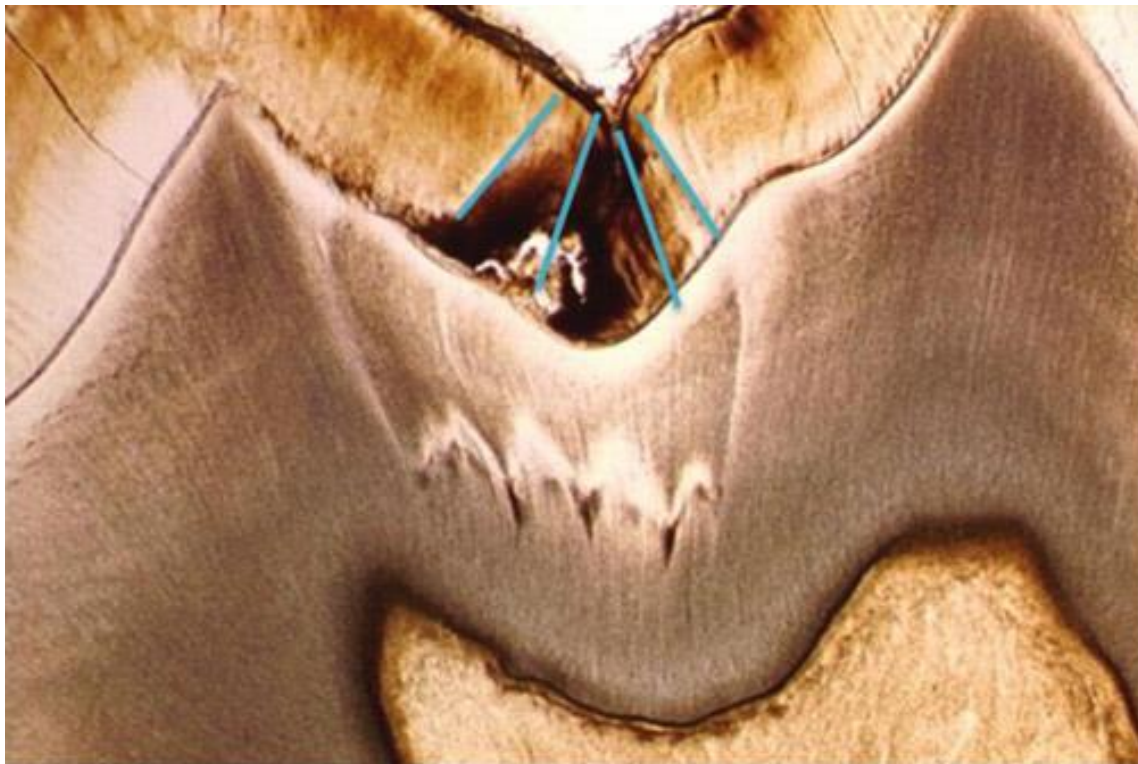


Histopathology of Dental Caries

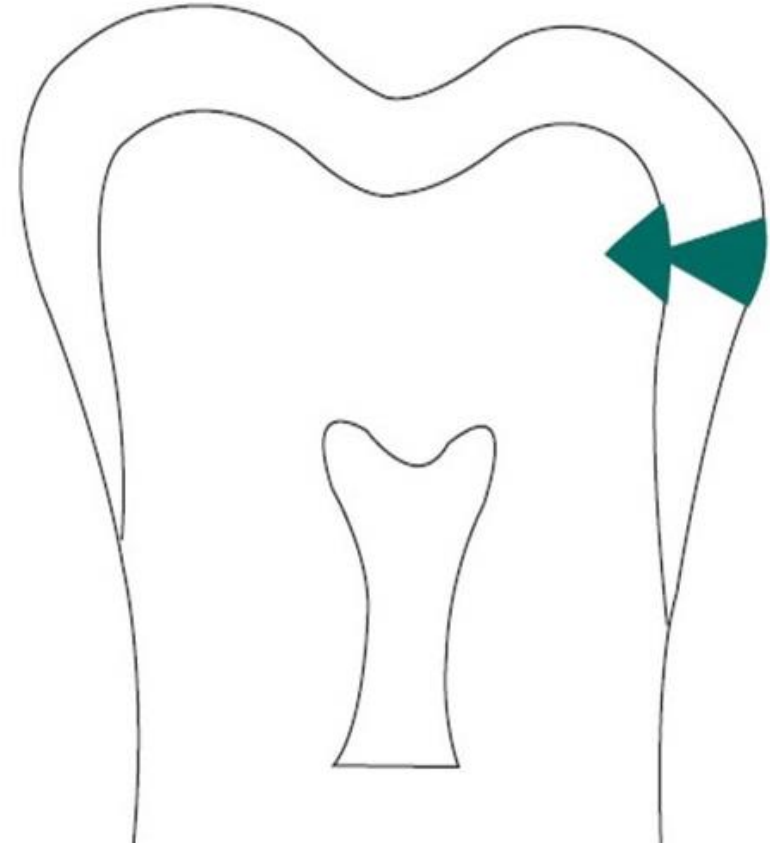
Cones/Triangles of Dental Caries



Pit and Fissure Caries

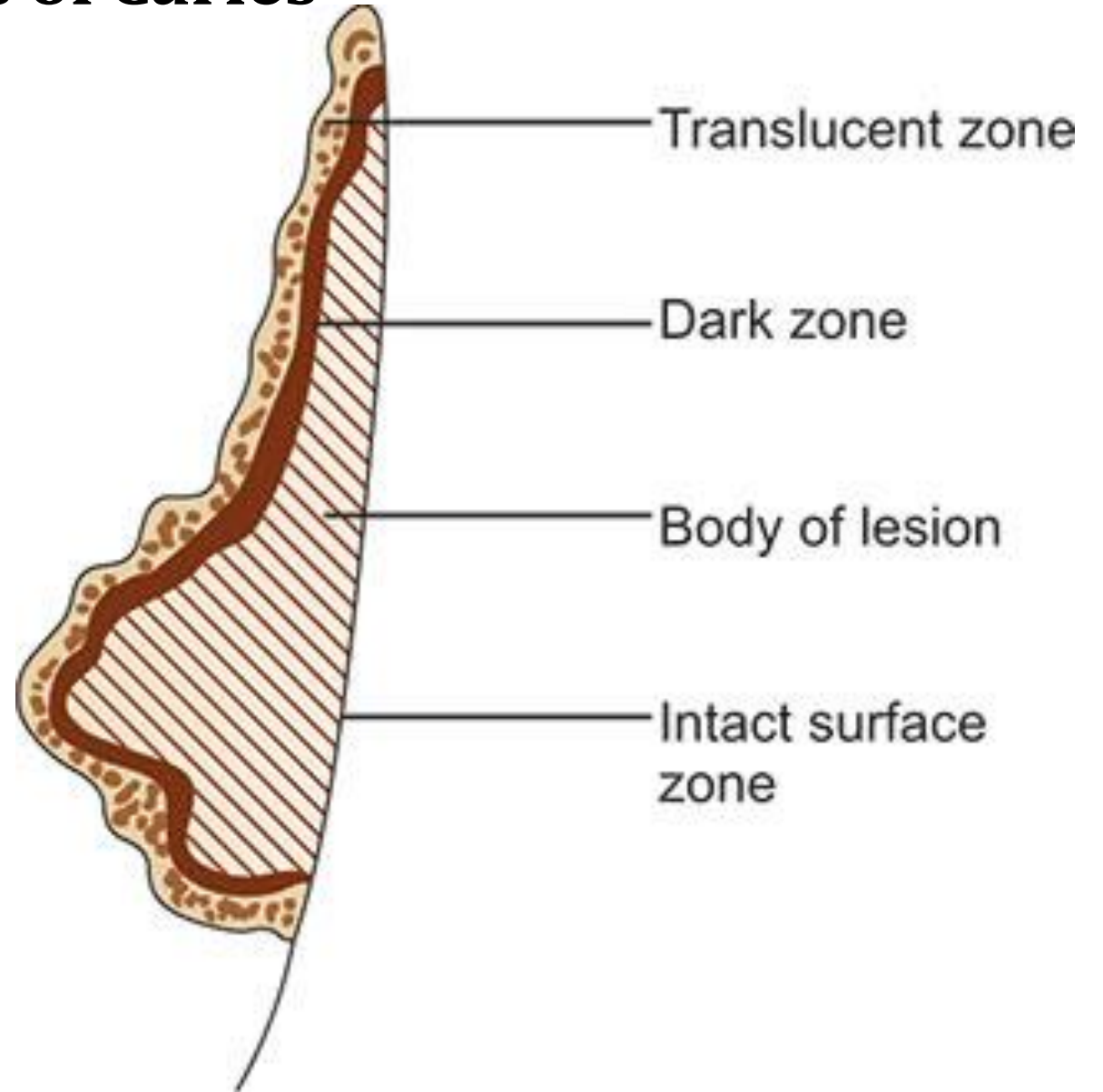


Smooth Surface Caries



Enamel Caries

Zones of Caries



The zones seen before complete disintegration of enamel are:

Zone 1: Translucent zone,

- lies at the advancing front of the lesion,
- slightly more porous than sound enamel,
- it is not always present

Zone 2: Dark zone,

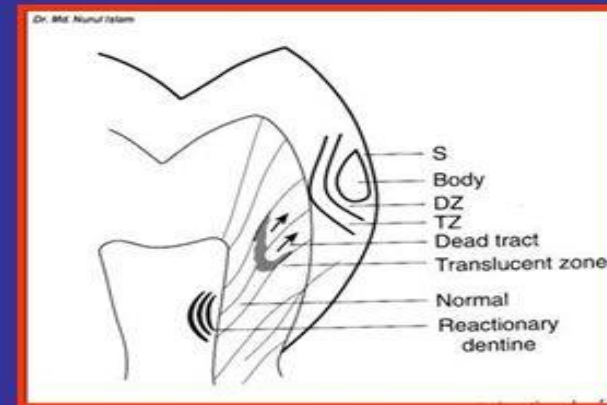
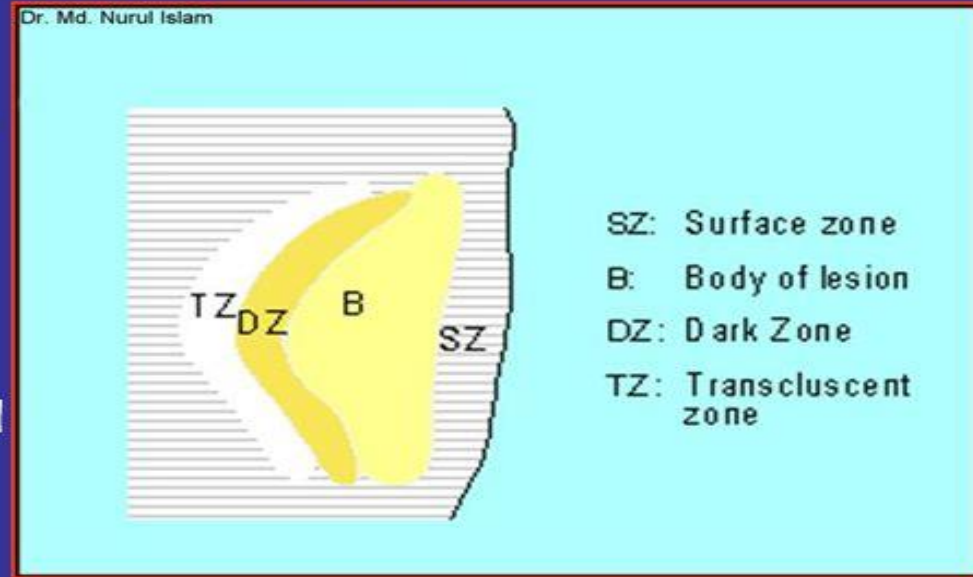
- this zone is usually present and referred to as positive zone
- formed due to demineralization.

Zone 3: Body of the lesion,

- found between the surface and the dark zone,
- it is the area of greatest demineralization,

Zone 4: Surface zone,

- relatively unaffected area,
- greater resistance probably due to greater degree of mineralization and greater F concentration.



Dentin Caries

Zones of Caries

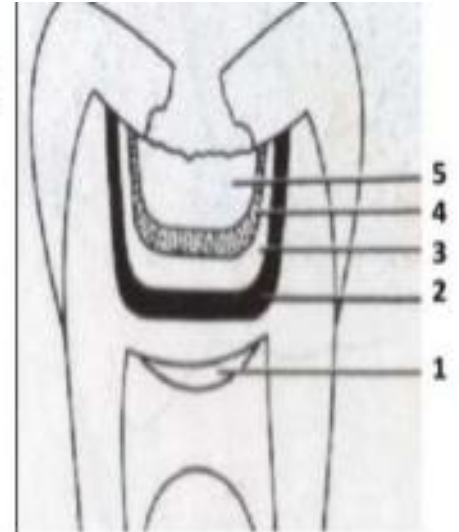
Zone 1: Zone of **F**atty degeneration

Zone 2: Zone of dentinal **S**clerosis

Zone 3: Zone of **D**ecalcification

Zone 4: Zone of **B**acterial invasion

Zone 5: Zone of **D**ecomposed dentin



Zone – 1: Fatty Degeneration

- This zone represents the innermost layer of the carious dentine
- Dentinal tubules appear normal.
- No crystals in the lumen of the tubules.
- No bacteria in the tubules.

Zone – 2: Dentinal Sclerosis

- This is the zone of dentinal sclerosis
- It is characterized by the deposition of very fine crystal structures within the dentinal tubules at the advancing front.

Zone -3: Decalcification

- This zone appears transparent and this is because of the demineralization of dentin due to caries.
- It is softer than normal dentin, further loss of mineral ions from inter tubular dentin.

Zone-4: Bacterial Invasion

- This zone is called the turbid dentin and is marked by the widening and distortion of the dentinal tubules ,which are packed with microorganism.
- Little amount of mineral present and moreover denaturation of collagen fibres also takes place.

Zone – 5 Decomposed Dentin

- This is outermost zone characterized by complete destruction of the dentinal tubules.
- The areas of decomposition of dentin that occur along the direction of the dentinal tubules called liquefaction foci of miller.
- The area perpendicular to the direction of the dentinal tubules are called the transverse clefts.
- Bacteria may no longer remain within dentinal tubules and they destroy the peri-and inter-tubular dentin

Management of Dental caries

Management of Dental caries

I. Prevention of Dental Caries

II. Restorative Treatment of Dental Caries

I. Prevention of Dental Caries

1. Oral Hygiene (Mechanical and Chemical)
2. Diet Modification
3. Fluorides
4. Pit and Fissure Sealants
5. Salivary Stimulants
6. Advanced preventive methods (Lasers and Polymeric coatings)

1. Oral Hygiene (Mechanical and Chemical)



Brushing



Flossing



Prophylaxis

1. Oral Hygiene (Mechanical and Chemical)



Mouthwash



Dentrifice



Disclosing agents (Erythrosine)

2. Diet Modification



Fibrous food



Cheese



Tea



?? Artificial Sweeteners ??



Milk

3. Fluorides

Systemic



Topical



4. Pit and Fissure Sealants



**Before Fissure
Sealant**

**After Fissure
Sealant**

Step by Step Fissure Sealants



Tooth Without Sealant



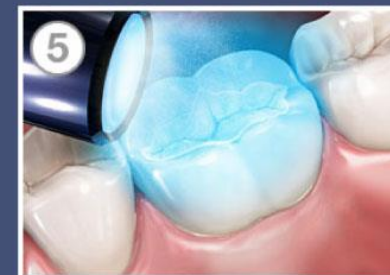
Tooth Cleaned and Dried



Etching Solution is Applied



Liquid Sealant is Applied



Curing Light Hardens Sealant



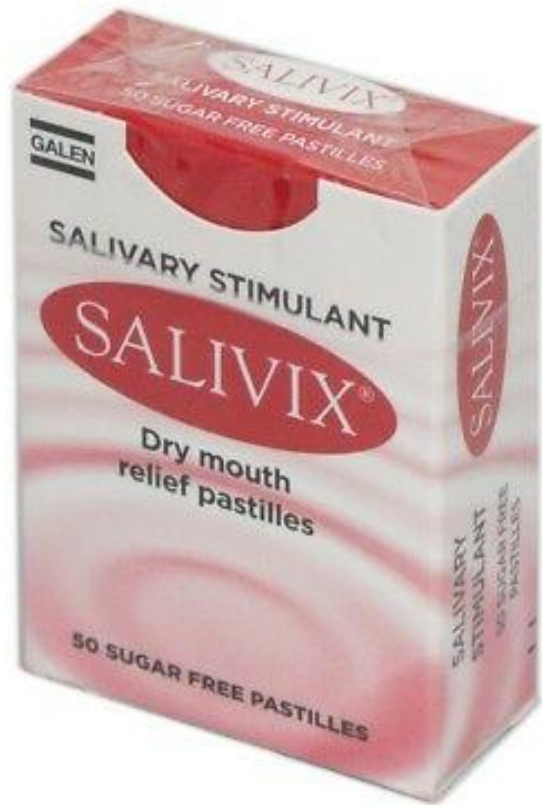
Final Sealant

5. Salivary Stimulants



Type of cause	Cause
Iatrogenic	Drugs
	Local radiation
	Chemotherapy
	Chronic graft versus host disease
Developmental causes	Salivary gland agenesis / atresia
Pathologic causes	Sialolithiasis
	Sialadenitis
	Sjogren's syndrome
Diseases of salivary glands	HIV infection
Infections	Hepatitis C virus infection
Metabolic disorders	Diabetes mellitus
Others	Amyloidosis
	Hemochromatosis
	Wegener's disease

5. Salivary Stimulants

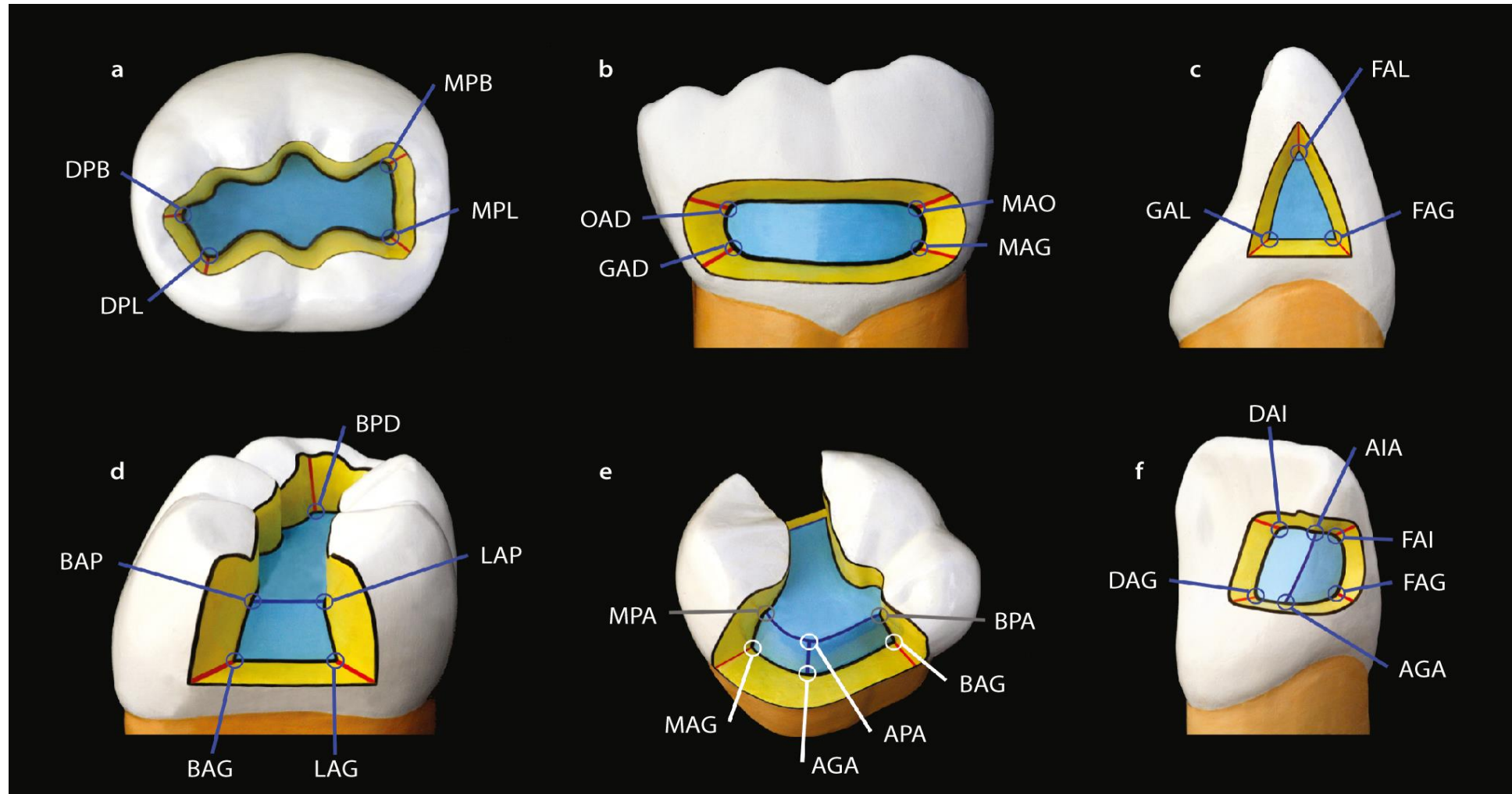


II. Operative Treatment of Dental Caries

>>>OPERATIVE DENTISTRY COURSES<<<

- **Preclinical Operative (3 BDS)**
- **Clinical Operative Dentistry - 1 (4BDS)**
- **Clinical Operative Dentistry - 2 (5BDS)**
- **Comprehensive Care Clinics (6 BDS)**

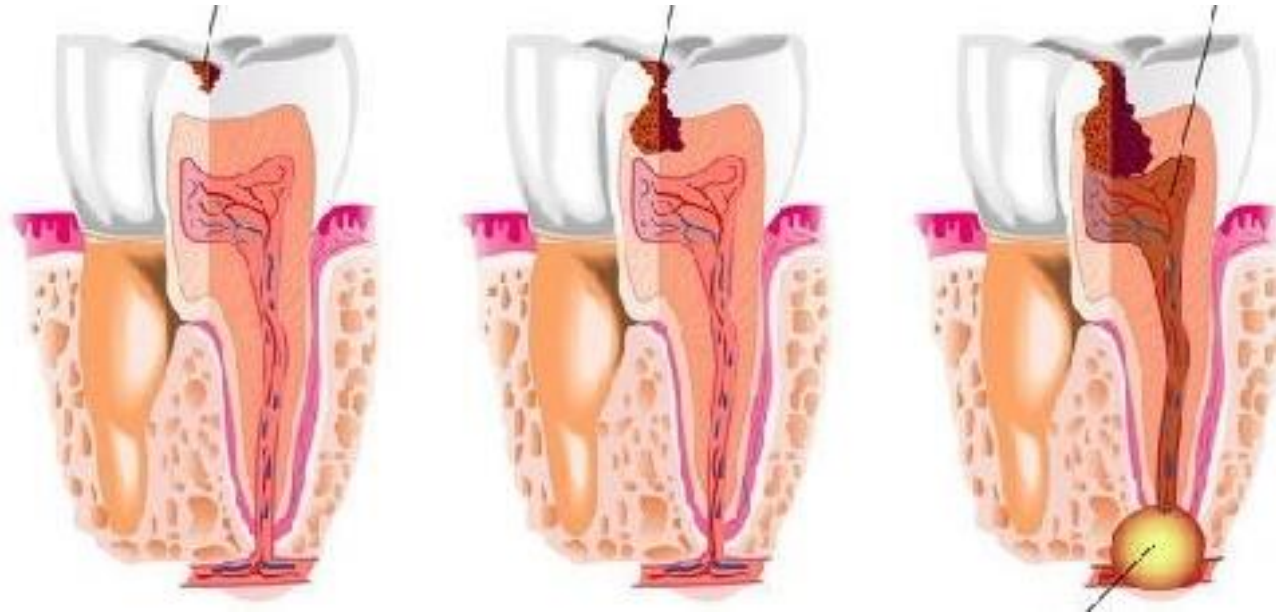
Teeth Preparations and Restorations



Teeth Preparations and Restorations



Sequela of Dental caries



Sequela of Dental Caries and Spread of Oral Infection

Sequela of Dental Caries

PATHOPHYSIOLOGY

- 1 Localised destruction of enamel by acid produced by bacterial metabolism of (refined) carbohydrates
- 2 Progressive bacterial invasion and destruction of underlying dentine
- 3 Mild inflammation of the underlying dental pulp in response to toxic products diffusing from carious dentine, i.e. pulpitis
- 4 Bacterial invasion of dental pulp causing severe pulpitis

Lower molar teeth in sagittal section



SYMPTOMS

Asymptomatic

Asymptomatic

Tooth abnormally sensitive to temperature changes, especially cold, and sweet foods

Constant dull aching pain
Increased sensitivity to heat

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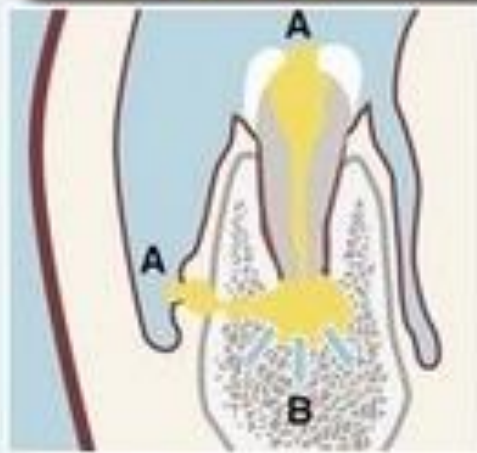
5 Necrosis of dental pulp
(often sudden)

6 Inflammation of periapical
area and formation of a
periapical (dental)
abscess

7 A Spontaneous drainage
of abscess into mouth via
exposed pulp canal or
through supporting
alveolar bone

or

7 B Chronic periapical
abscess formation with
increasing bone resorption



Molar tooth in coronal section

Pulpitic pain resolves

Dull aching pain; tooth
sensitive to pressure and
percussion; increasing
pain and localised swelling

Pain and swelling resolve;
'gum-boil' may appear and
discharge intermittently

Occasional bouts of aching
pain in jaw

Summary: Sequelae of Dental Caries



Spread of Oral Infection

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graph TD; A[Spread of Oral Infection] --> B[Cellulitis]; A --> C[Tissues Spaces]; A --> D[Ludwig's Angina]; A --> E[Maxillary Sinusitis]; A --> F[Focal Infection];
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Cellulitis

Tissues Spaces

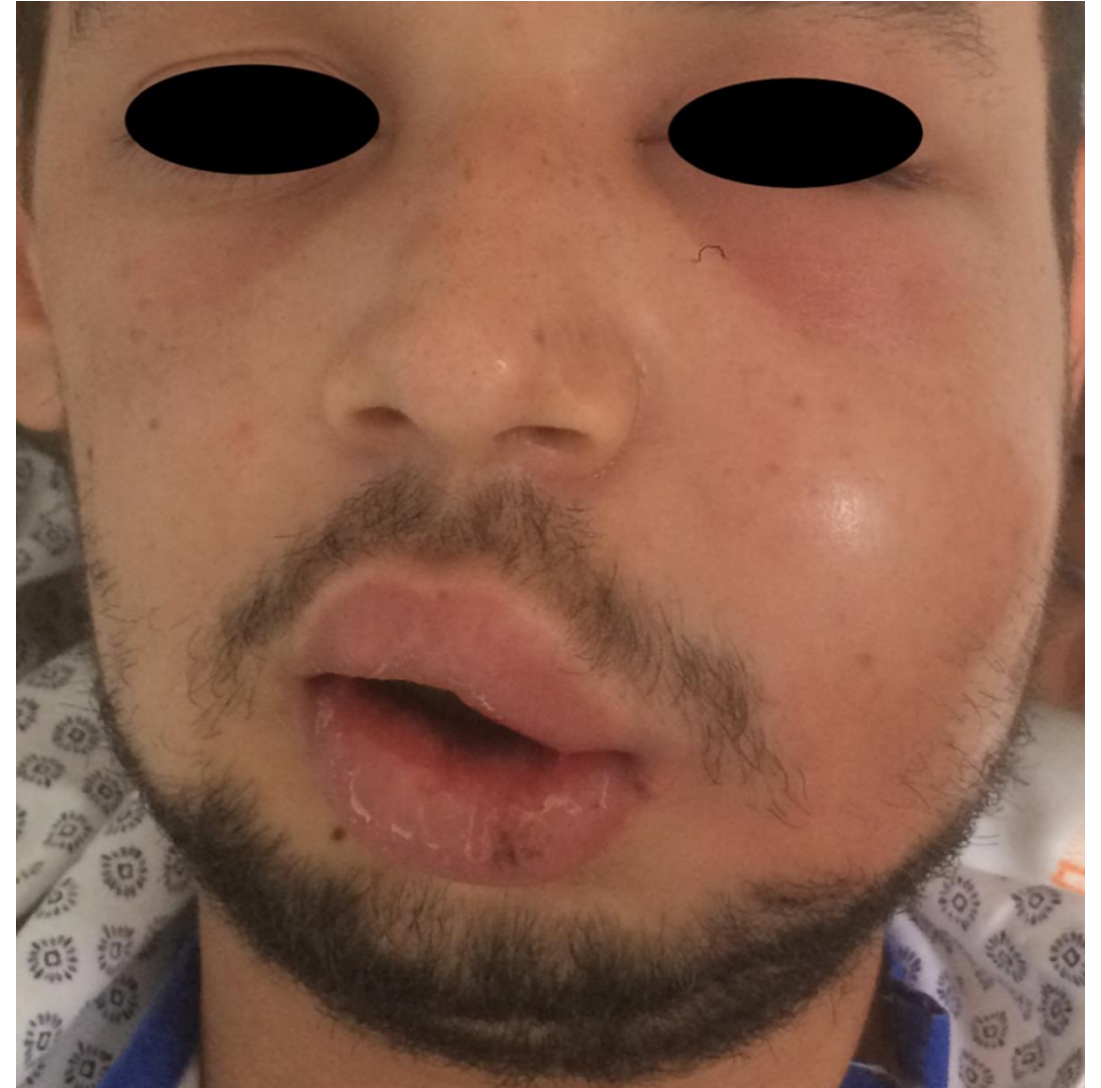
Ludwig's Angina

Maxillary
Sinusitis

Focal
Infection

Cellulitis

Diffuse inflammation of soft tissues which is not circumscribed or confined to one area and tends to spread through tissue spaces and along facial planes.



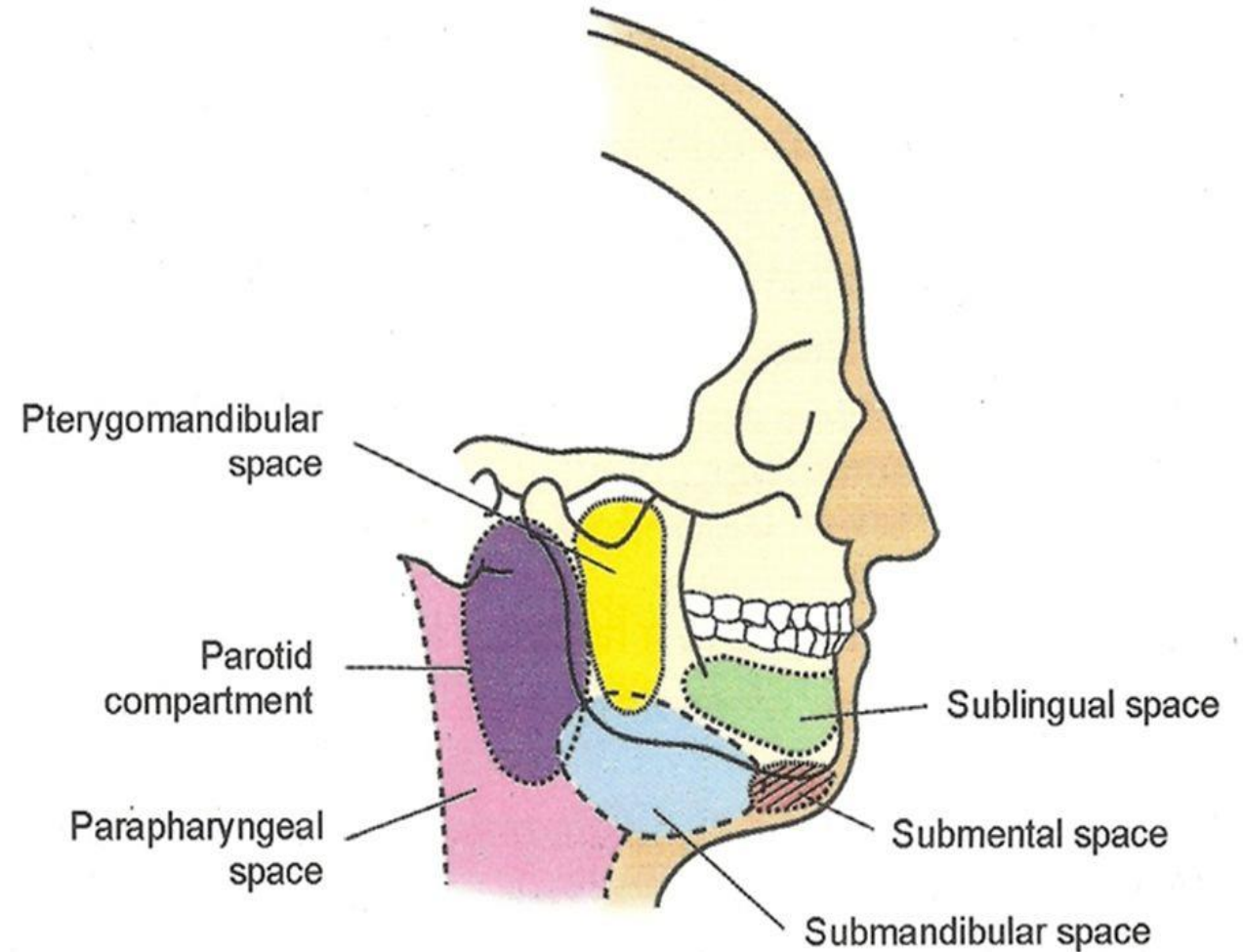
Signs and Symptoms of Abscess and Cellulitis²¹

SIGN OR SYMPTOM	ABSCESS	CELLULITIS
Localization	Defined	Diffuse
Size	Small	Large
Appearance	Peripherally reddened	Reddened
Pain	Moderate	Severe
Surface temperature	Hot	Moderately elevated
Tissue fluid	Pus	Serosanguinous (pus)
Bacteria	Anaerobic	Mixed
Palpation	Fluctuant	Hard and very tender



Tissues Spaces

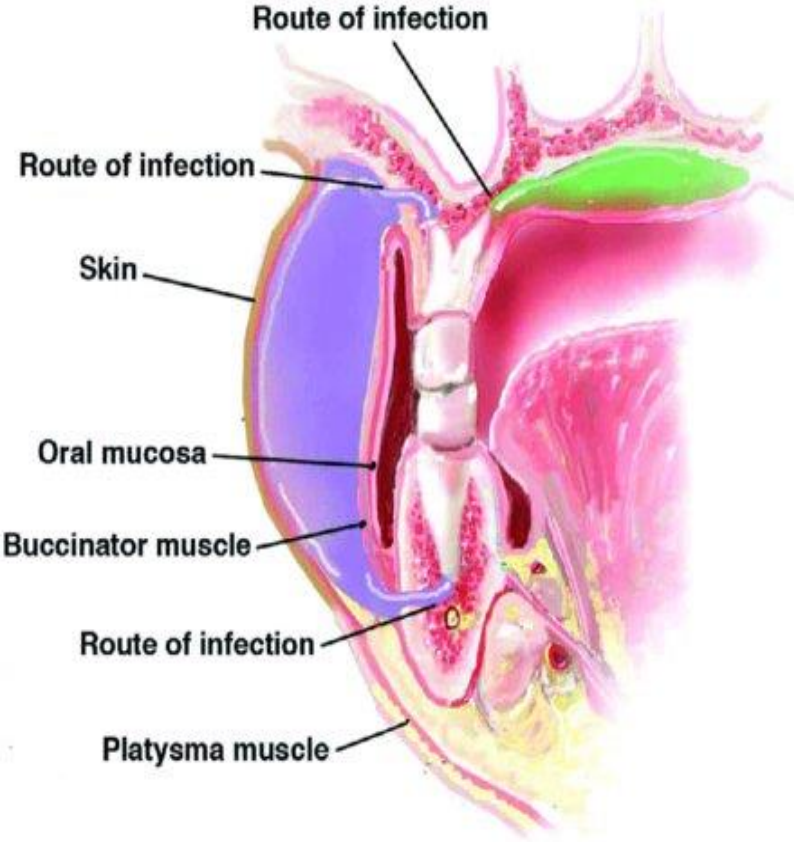
Potential spaces situated
between the planes of fascia
that form natural pathways
along which infection may
spread and become localized



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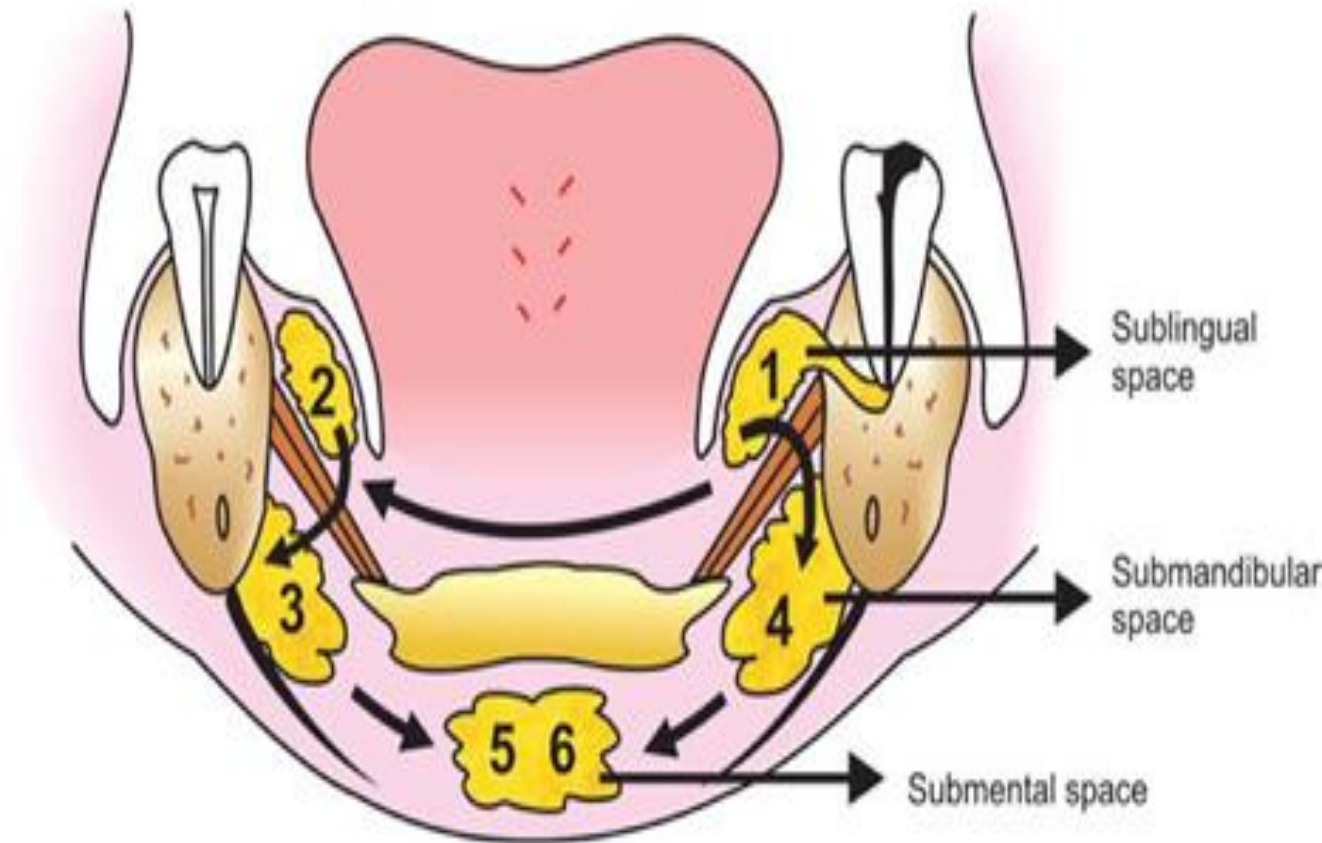


Buccal and Palatal Spaces

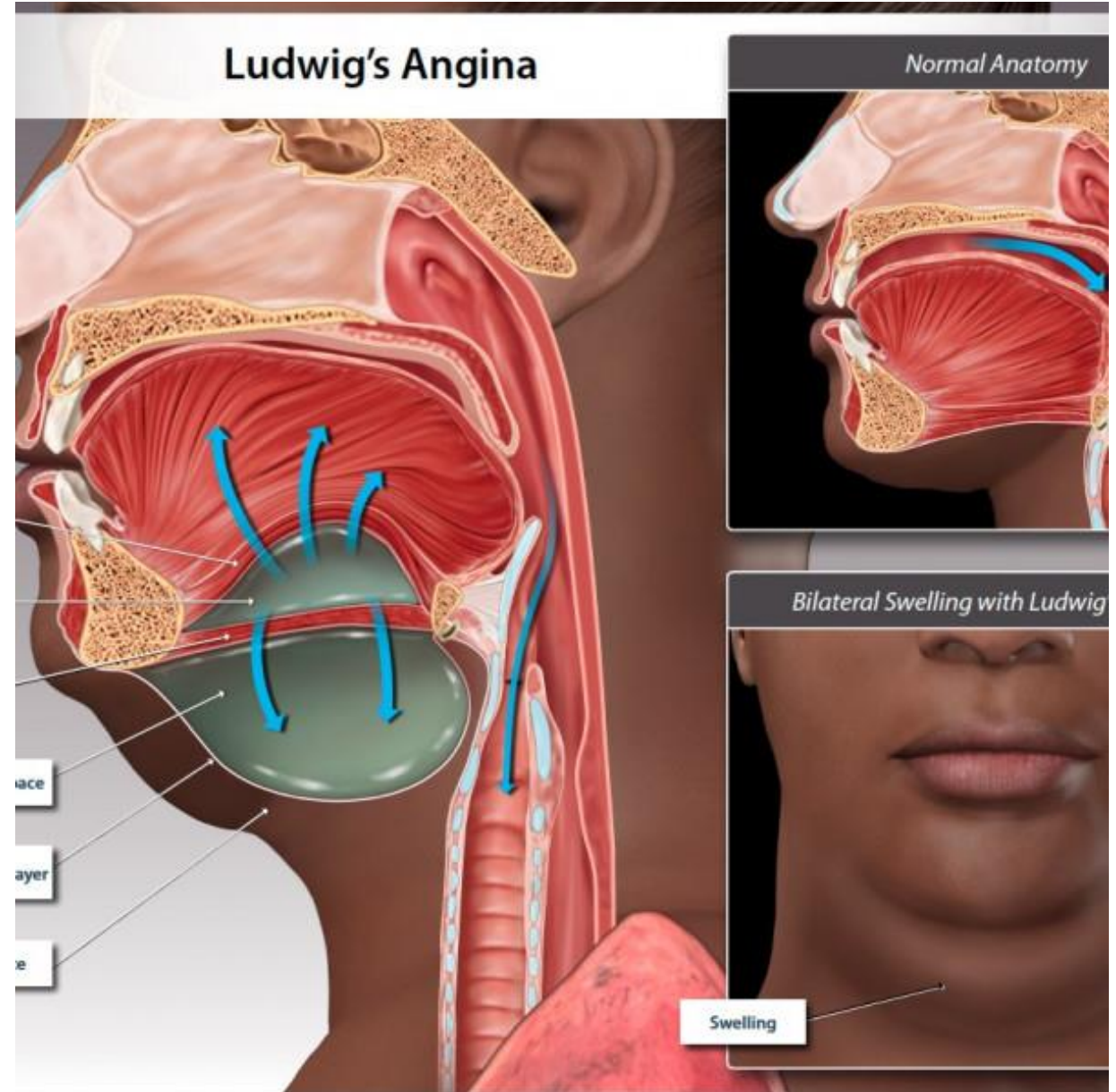


Ludwig's Angina

Acute potentially life threatening toxic cellulitis beginning usually in submandibular space and secondarily involving the sublingual and submental spaces.

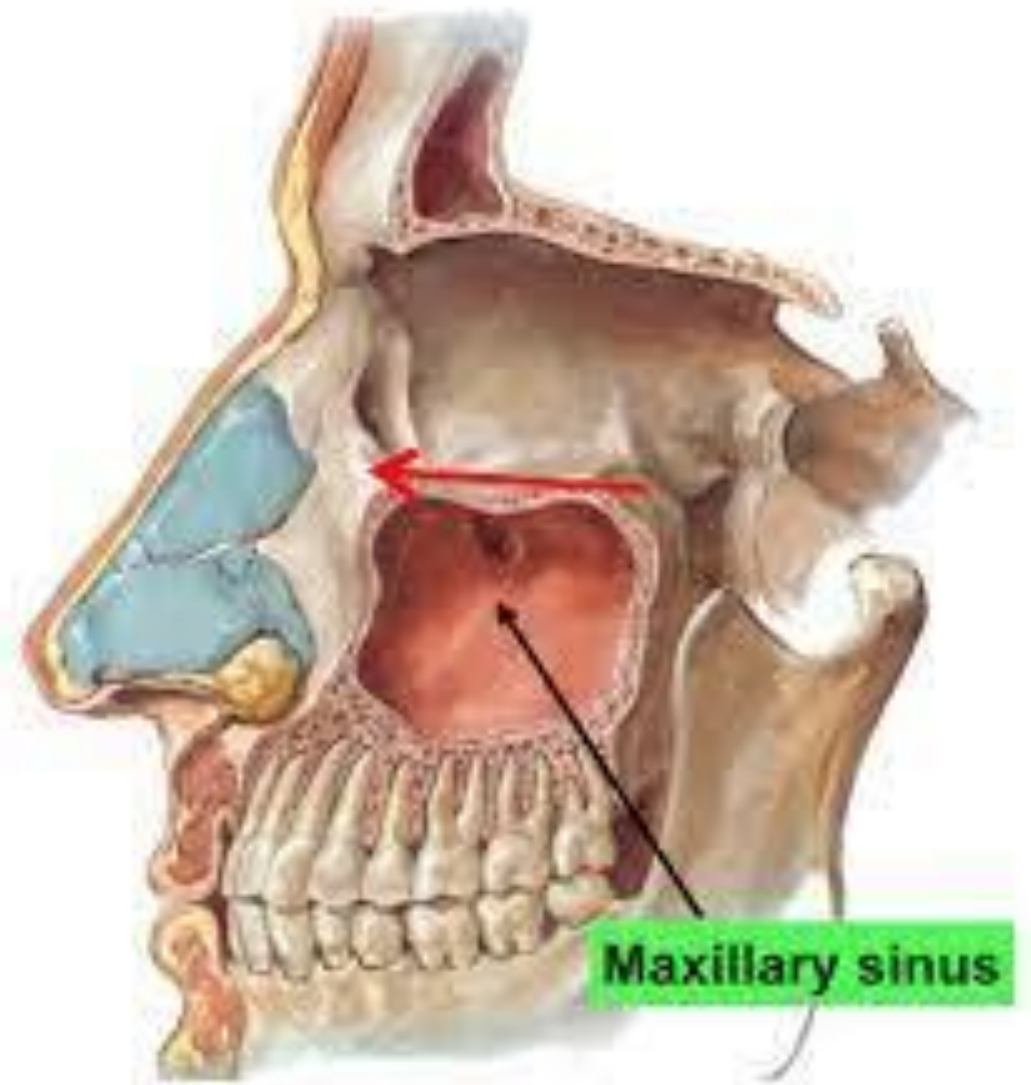


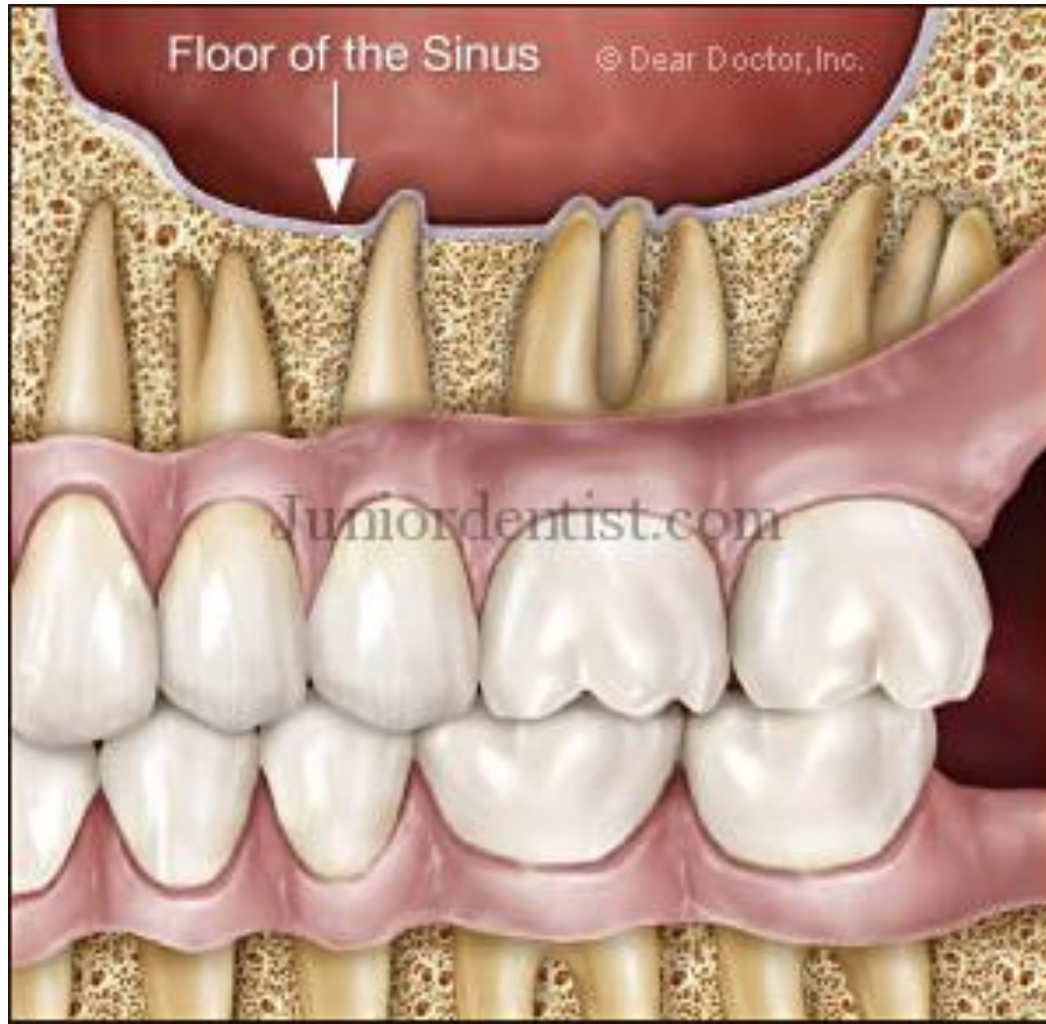
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Maxillary Sinusitis

- Acute or chronic inflammation of maxillary sinus is often due to direct extension of dental infection, but also originates from infectious diseases due to bacteria fungus or virus.





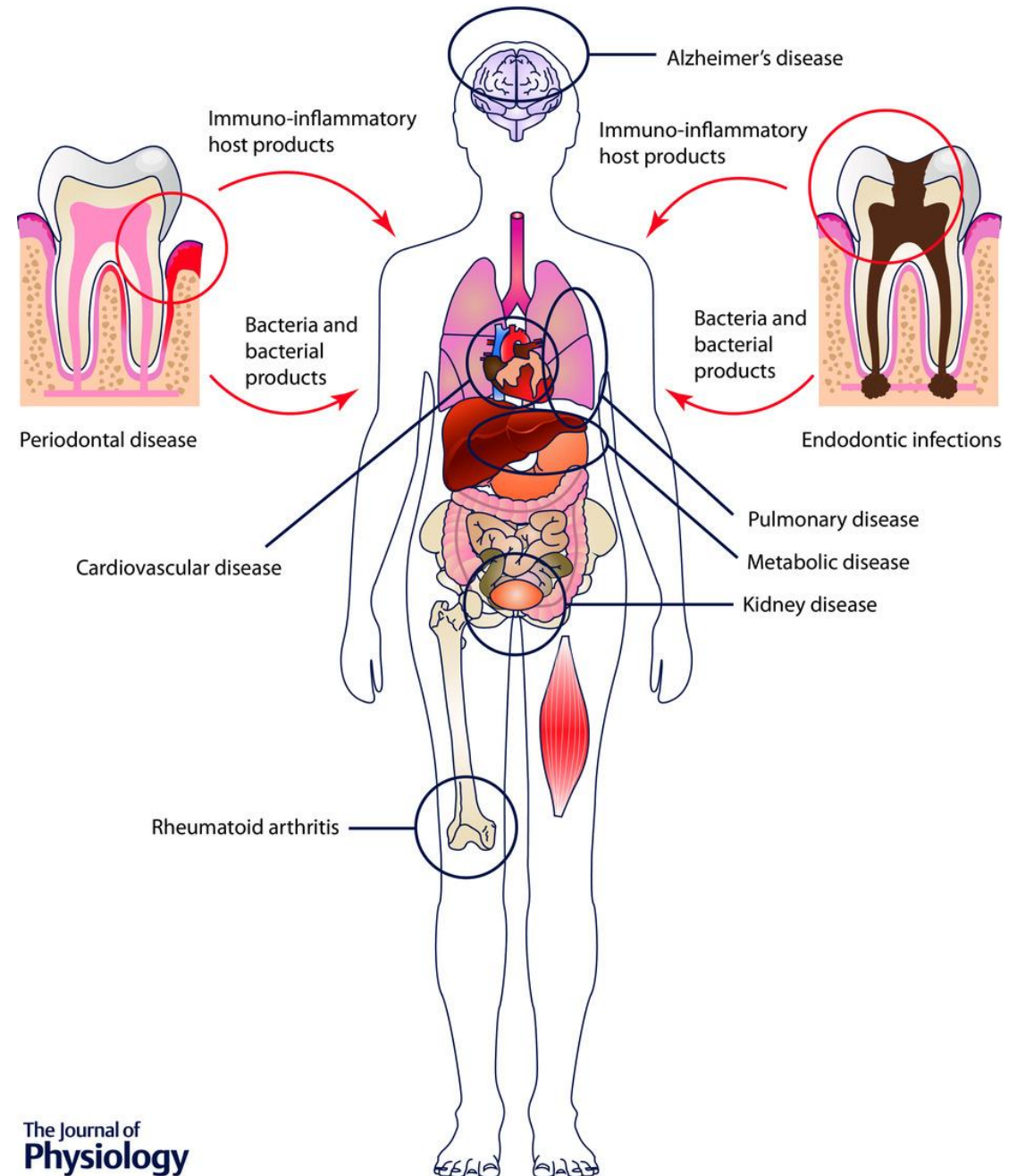
Normal Anatomy



Periapical disease extending in sinus

Focal Infection

- A **focal infection** is a localized or general infection caused by the dissemination of microorganisms or toxic products from a **focus of infection**.



Summary of Sequelae of Dental Caries



Portfolio Activity

Dental Caries

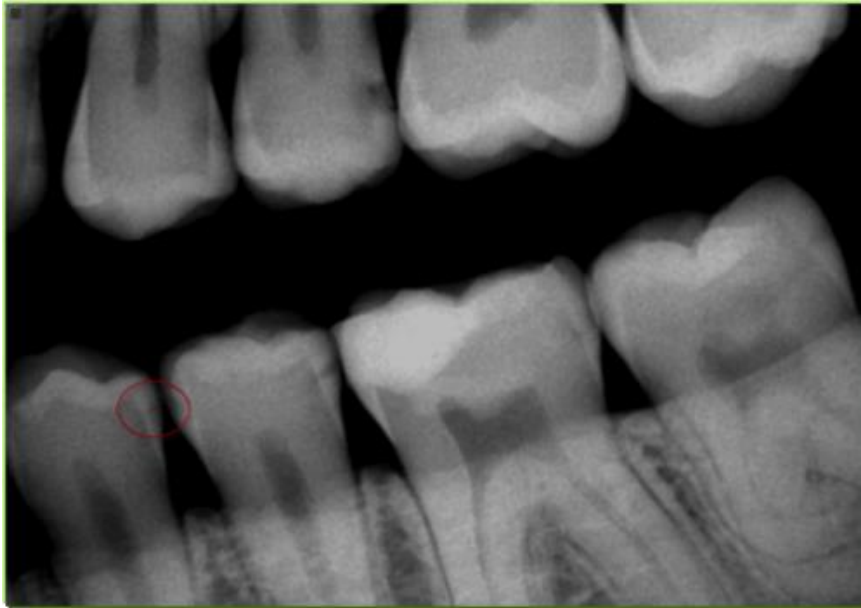
CASE # 4

Clinical Presentation, Key Features & Diagnosis



CASE # 5

Clinical Presentation, Key Features & Diagnosis



Essential Reference

- Shafer's text book of oral pathology – 2020 Elsevier (9th edition).

Chapter 11: Dental caries : 369 to 403

Chapter – 14: Spread of Oral Infection Page: 452 to 470

Thank You